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CS610P - Computer Networks (Practical)

Final term Sample Paper

1. Which one of the following addresses is used to uniquely identify each device in a TCP/IP network?
 - a) TCP
 - b) IP
 - c) FTP
 - d) UDP
2. _____ is the correct IP address.
 - a) 192.168.0.256
 - b) 10.25.0.300
 - c) 192.168.0.250
 - d) 172.256.10.2
3. Which of the following cable is used to connect a PC with a switch?
 - a) Straight-through
 - b) Crossover
 - c) Roll over
 - d) Console
4. The maximum length for thick coaxial cable may be up to _____.
 - a) 100 meters
 - b) 600 meters
 - c) 500 meters
 - d) 1000 meters
5. Which of the following connectors is used with twisted pair cable?
 - a) RJ-11
 - b) RJ-45
 - c) RJ-55
 - d) RJ-65

6. _____ device can analyze incoming data packets.
- a) Bridge
 - b) HUB
 - c) Repeater
 - d) Amplifier
7. _____ device is used to connect two different types of computer networks.
- a) Hub
 - b) Switch
 - c) Router
 - d) Repeater
8. Which of the following types of protocol determines the path of a packet through an inter-network?
- a) Routing
 - b) Rolling
 - c) Rated
 - d) Relevant
9. Suppose if you have large network with so many devices then _____ routing will be preferred.
- a) Auto
 - b) Static
 - c) Dynamic
 - d) Straight
10. _____ allows users to capture and analyze the traffic running on a computer network.
- a) Wire shark
 - b) Packet tracer
 - c) IDS
 - d) Router
11. Which of the following protocol used to translate an IP address to a hardware address?

a) ARP

b) DNS

c) FTP

d) TCP

12. IP address has _____ different classes.

a) 3

b) 4

c) 5 (A, B, C, D, E)

d) 6

13. _____ bits are used in Class C for network portion.

a) 8

b) 16

c) 24

d) 32

14. Which of the following number of bits will be used for host portion in class A?

a) 8

b) 16

c) 24

d) 32

15. Which of the following address belongs to class A?

a) 50.20.1.0

b) 136.25.2.10

c) 199.3.2.1

d) 236.12.0.1

16. Which of the following address belongs to class C?

a) 200.40.1.2

b) 10.25.3.0

c) 172.16.2.1

d) 235.2.1.0

17. Which of the following is a loop back address?

- a) 127.0.0.1
- b) 192.168.2.5
- c) 172.16.2.3
- d) 10.0.1.2

18. Which of the following is a private IP address?

- a) 192.168.2.1
- b) 224.25.1.2
- c) 172.85.2.1
- d) 89.2.5.1

19. Suppose if we borrow 3 bits from host portion in class C then _____ subnets and _____ usable hosts per subnet will be created.

- a) 4 and 62
- b) 2 and 126
- c) 8 and 30
- d) 16 and 14

20. Identify the correct subnet mask for class C.

- a) 255.0.0.0
- b) 255.255.0.0
- c) 255.255.255.0
- d) 255.255.255.255

21. Which of the following is a multicast address?

- a) 224.0.2.1
- b) 192.168.2.1
- c) 173.16.52.1
- d) 10.25.4.0

22. _____ is the correct subnet mask for the network address 192.29.0.0 to get 10 subnets?

- a) 255.255.255.128
- b) 255.255.255.192
- c) 255.255.255.240
- d) 255.255.255.224

23. Which of the following is a correct CIDR address notation?

- a) 192.168.1.1/24
- b) 192.168.1.1\24
- c) 192.168.1.1-24
- d) 192.168.1.1!24

24. What is the purpose of the slash notation in CIDR addressing?

- a) To indicate the number of IP addresses in a subnet
- b) To indicate the network address of a subnet
- c) To indicate the broadcast address of a subnet
- d) To indicate the subnet mask of a subnet

25. Each value in IPV6 address is separated by _____ sign.

- a) Dolor sign
- b) Colon
- c) Percentage
- d) Asterisk

Question No: 21 (Marks: 05)

Suppose a currency exchange company wants to design a computer network. The company has its head office at Lahore and recently opened a branch in Faisalabad. The head office comprising of 3 floors that will be connected with Lahore branch. Employees in Lahore are in need to access the confidential information from the server located in Lahore. By keeping in mind the scenario, You are required to answer the following questions:

In Lahore branch, which type of intelligent networking device will be used to connect computers with each other?

1. Name the address that will be used for communication among the devices in LAN of Lahore.

Intelligent networking device for connecting computers in Lahore branch: Switch

- 2. Which secure connection will be created for communication between Lahore and head office computers?**

Address used for communication among devices in LAN of Lahore:
IP address (e.g., 192.168.x.x)

- 3. Name the networking device that will be used in head office to connect all three LANs with each other.**

Secure connection between Lahore and head office computers: VPN
(Virtual Private Network)

- 4. Which networking device will be used for creating connection between Lahore and Faisalabad office?**

Networking device in head office to connect all LANs: Router

Question No: 22 (Marks: 05)

The company ABC has its main office in Karachi and two smaller branches in different locations. Each branch has its own LAN with 100 computers. The main office has 300 computers connected to its LAN. The LANs of each branch are connected to the head office using Wide Area Network (WAN), which allows the branches to communicate and share resources with the head office. Furthermore, to ensure the connectivity of devices in all LANs, the network administrator regularly performs tests. After reading scenario you are required to answer the following questions:

- 1. Which Type of address will be used for communication among devices in a LAN.**

Address used for communication among devices in a LAN: IP address

- 2. Which type of address will be used for communication between different branches.**

Address used for communication between different branches: WAN IP address

- 3. Which type of command will be used to ensure the connectivity?**

Command used to ensure connectivity: Ping command

Question No: 23 (Marks: 05)

How many minimum IP addresses will be purchased by the ISP to accommodate 450 customers assuming that ISP can only purchase IP addresses range as a block of 2^n . Suppose in future, ISP wants to extend its service in small colony to accommodate more customers, then how many customers will be accommodated with the remaining available addresses after accommodating 450 customers?

- Minimum IP addresses purchased by ISP to accommodate 450 customers: 512 IP addresses (since $2^9 = 512$)
- Remaining available addresses after accommodating 450 customers: 62 IP addresses (since $512 - 450 = 62$)

Question No: 24 (Marks: 05)

Considering the following IP address and its corresponding subnet mask:

IP Address: 192.18.5.202

Subnet Mask: 255.255.255.248

The no. of subnet bits determined by subnet mask 255.255.255.248 which has 5 bits in the last octet, subtracting 24 gives us 3 bits for subnetting.

Total no. of subnets is determined by ($2^3 = 8$)

Total no. of useable hosts per subnet is determined by ($2^3 - 2 = 6$, excluding network and broadcast addresses)

No. of subnet bits	3
Total No. of subnets	8
Total No. of useable hosts per subnet	6

Network address of this subnet	192.18.5.200
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Question No: 25 (Marks: 05)

Suppose you have designed an office network of two floors that are attached with each other using router device in the Packet Tracer Tool. You are required to write all steps which will be used to assign IP address and default gateway to a router device.

Steps to assign IP address and default gateway to a router device in Packet Tracer:

1. Enter privileged EXEC mode on the router: enable
2. Enter global configuration mode: configure terminal
3. Assign IP address to an interface: interface <interface_name>
Example: interface FastEthernet0/0 ip address <ip_address> <subnet_mask> Example: ip address 192.168.1.1 255.255.255.0
4. Set default gateway: ip default-gateway <gateway_ip> Example: ip default-gateway 192.168.1.254
5. Exit configuration mode: end
6. Save configuration: copy running-config startup-config

Question No: 26 (Marks: 05)

Identify the IP address class for each IP address in the following table:

Binary value of first octet	IP address class
10.25.3.56	Class A
224.25.21.0	Class D (Multicast)
198.125.100.1	Class C
75.45.23.1	Class A
245.56.89.20	Class D (Multicast)

Question No: 27 (Marks: 05)

Consider the IP address 215.35.5.1/25 and provide the following information:

Total number of subnets: 128 ($2^7 = 128$)

Total number of useable hosts: 126 ($2^7 - 2 = 126$, excluding network and broadcast addresses)

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